

1. At leading order in perturbation theory, two distinct Feynman graphs can be drawn, specifically the  $s$ - and  $u$ -channel, which can be seen in Fig. 1.

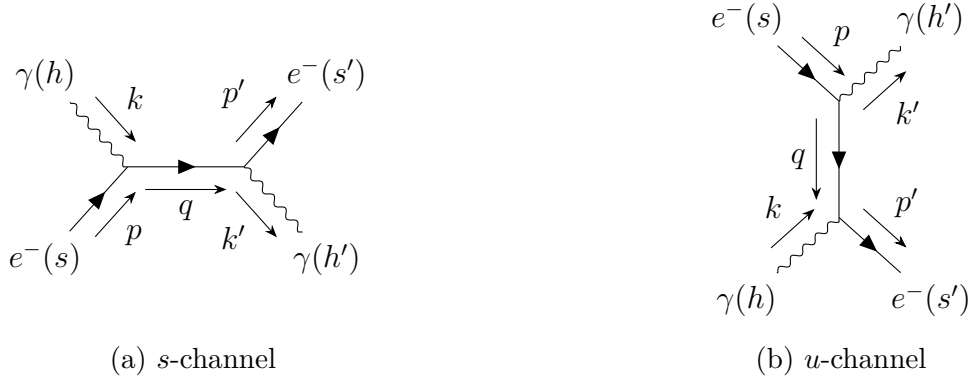


Figure 1: Feynman graphs for  $e^- + \gamma \rightarrow e^- + \gamma$  Compton scattering

By reading out the Feynman graphs, we obtain the amplitudes

$$\begin{aligned}
 i\mathcal{T}_s &= (-ie)^2 \bar{u}(p', s') \varepsilon_{h'}^{*\mu}(k') \gamma_\mu i \frac{\gamma \cdot (p + k) + m}{s - m^2 + i0} \varepsilon_h^\nu(k) \gamma_\nu u(p, s) \\
 i\mathcal{T}_u &= (-ie)^2 \bar{u}(p', s') \varepsilon_h^\mu(k) \gamma_\mu i \frac{\gamma \cdot (p - k') + m}{u - m^2 + i0} \varepsilon_{h'}^{*\nu}(k') \gamma_\nu u(p, s) \\
 i\mathcal{T}_{fi} &= i\mathcal{T}_s + i\mathcal{T}_u
 \end{aligned}$$

2. Assuming the Lorentz gauge, we have  $\partial_\mu A^\mu = 0$ , which in the momentum space corresponds to  $k_\mu \varepsilon_h^\mu(k) = 0$ . Given the fact that the momentum vector  $k^\mu$  of a massless particle is null  $k_\mu k^\mu = 0$ , the transformation

$$\varepsilon_k^\mu(k) \rightarrow \varepsilon_k^\mu(k) + \alpha k^\mu,$$

where  $\alpha$  satisfies  $\square \alpha = 0$ , does not break the Lorentz gauge. Now we assume that  $p \cdot \varepsilon \neq 0$ . We can now take

$$\alpha = -\frac{p \cdot \varepsilon}{p \cdot k}.$$

Using the transformation of the field, we obtain

$$\varepsilon_k^\mu(k) \rightarrow \varepsilon_k^\mu(k) - \frac{p \cdot \varepsilon}{p \cdot k} k^\mu,$$

which trivially results in

$$p_\mu \left[ \varepsilon_k^\mu(k) - \frac{p \cdot \varepsilon}{p \cdot k} k^\mu \right] = 0.$$

In the  $x$ -space, the transformation has the form

$$A^\mu(x) \rightarrow A^\mu(x) + \partial^\mu \alpha(x),$$

where

$$\alpha(x) = \sum_h \int d\tilde{k} \left[ \frac{p \cdot \varepsilon_h(k)}{p \cdot k} a(k, h) e^{-ik \cdot x} + \text{h.c.} \right].$$

From this relation, we see that this ensures the desired orthogonality between  $p$  and  $\varepsilon$  for both  $k$  and  $k'$ . In addition, we see that  $\alpha$  satisfies  $\square \alpha = 0$  as the  $\square$  operator results in  $k_\mu k^\mu$  in the integrand, which is null.

**3.** In the rest-frame of the electron, we define the momenta of incoming particles as

$$p^\mu = (m, 0, 0, 0), \quad k^\mu = (\omega, 0, 0, \omega),$$

where we have chosen to set the  $z$ -axis as the direction of the incoming photon. Given the symmetry of the scattering process to rotation around the  $z$ -axis, we can choose the scattering plane as the  $xz$ -plane and parametrize the direction of the outgoing photon using scattering angle  $\vartheta$

$$k' = (\omega', \omega' \sin \vartheta, 0, \omega' \cos \vartheta).$$

From the conservation of energy and momentum, we have

$$p' = p + k - k' = (m + \omega - \omega', -\omega' \sin \vartheta, 0, \omega - \omega' \cos \vartheta).$$

From the on-shell condition for  $p'$  we obtain the relation for  $\omega'$

$$m^2 \stackrel{!}{=} (m + \omega - \omega')^2 - (\omega' \sin \vartheta)^2 - (\omega - \omega' \cos \vartheta)^2 \implies \omega' = \frac{m\omega}{m + \omega(1 - \cos \vartheta)}$$

Using this parametrization, the expression

$$\frac{1}{2} \sum_{s, s'} |\mathcal{T}_{fi}|^2 = e^4 \left[ \frac{p \cdot k'}{p \cdot k} + \frac{p \cdot k}{p \cdot k'} + 4 [\varepsilon(k) \cdot \varepsilon(k')]^2 - 2 \right]$$

can be simplified to

$$\frac{1}{2} \sum_{s, s'} |\mathcal{T}_{fi}|^2 = e^4 \left[ \frac{\omega'}{\omega} + \frac{\omega}{\omega'} + 4 [\varepsilon(k) \cdot \varepsilon(k')]^2 - 2 \right]$$

As for the polarization vectors, for the incoming photon we have the standard polarization according to the  $z$ -axis

$$\varepsilon_\pm(k) = \mp \frac{1}{\sqrt{2}} (e_1 \pm i e_2).$$

However, for the outgoing photon, the polarization axis is rotated in the  $xz$ -plane, resulting in

$$e'_1 = (0, \cos \vartheta, 0, -\sin \vartheta), \quad e'_2 = (0, 0, 1, 0) = e_2$$

which gives us the polarization vectors

$$\varepsilon_{\pm}(k') = \mp \frac{1}{\sqrt{2}}(e'_1 \pm ie'_2).$$

Their scalar product is then equal to

$$\varepsilon_h(k) \cdot \varepsilon_{h'}(k') = \cos^2 \frac{\vartheta}{2} \delta_{hh'} + \sin^2 \frac{\vartheta}{2} (1 - \delta_{hh'}).$$

[4.] First we derive the relation between the differential cross-section and the amplitude. We have

$$d\sigma = \frac{1}{j_{\text{flux}}} |\mathcal{T}_{fi}|^2 d\text{LIPS}_2,$$

where

$$j_{\text{flux}} = 4p \cdot k = 4m\omega$$

and

$$\begin{aligned} d\text{LIPS}_2 &= \frac{d^3 \mathbf{k}'}{(2\pi)^3 2\omega'} \frac{d^3 \mathbf{p}'}{(2\pi)^3 2E'} (2\pi)^4 \delta^{(4)}(k + p - k' - p') \\ &= \frac{d^3 \mathbf{k}'}{(2\pi)^3 2\omega'} \frac{d^3 \mathbf{p}'}{(2\pi)^3 2E'} (2\pi)^4 \delta^{(3)}(\mathbf{k} + \mathbf{p} - \mathbf{k}' - \mathbf{p}') \delta(\omega + m - \omega' - E') \\ &= \frac{d^3 \mathbf{k}'}{(2\pi)^2 2\omega'} \frac{1}{2E'} \delta(\omega + m - \omega' - E') \\ &= \frac{\omega'^2 d\omega'}{16\pi^2 \omega' E'} \delta(\omega + m - \omega' - E') d\Omega \\ &= \frac{\omega' d\omega'}{16\pi^2} \frac{1}{E'(\omega')} \delta(\omega + m - \omega' - E'(\omega')) d\Omega, \end{aligned}$$

where

$$E'(\omega') = \sqrt{m^2 + \omega'^2 + \omega'^2 - 2\omega\omega' \cos \vartheta}.$$

Using the relation

$$\delta(f(\omega')) = \frac{1}{|f'(\omega^*)|} \delta(\omega' - \omega^*),$$

where  $\omega^*$  is the root of  $f(\omega')$ , we again retrieve the already obtained relation for

$$\omega^* \equiv \omega' = \frac{\omega m}{m + \omega(1 - \cos \vartheta)}.$$

We then substitute for

$$|f'(\omega^*)| = \left| -1 - \frac{dE'(\omega')}{d\omega'} \right|_{\omega'=\omega^*} = \left| -1 - \frac{\omega^* - \omega^* \cos \vartheta}{E'(\omega^*)} \right| = \frac{m\omega}{\omega^* E'(\omega^*)},$$

where we have used the relation

$$E'(\omega^*) = \omega + m - \omega^*.$$

Overall, we obtain

$$d\text{LIPS}_2 = \frac{\omega'^2}{16\pi^2 m \omega} d\Omega \implies \frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 m^2} \frac{\omega'^2}{\omega^2} |\mathcal{T}_{fi}|^2.$$

By substituting the relation for the amplitude, we obtain

$$\left( \frac{d\sigma}{d\Omega} \right)_{hh'} = \frac{e^4}{64\pi^2 m^2} \frac{\omega'^2}{\omega^2} \left[ \frac{\omega'}{\omega} + \frac{\omega}{\omega'} + 4 \left[ \cos^2 \frac{\vartheta}{2} \delta_{hh'} + \sin^2 \frac{\vartheta}{2} (1 - \delta_{hh'}) \right]^2 - 2 \right].$$

After averaging the differential cross-section over incoming-photon helicities and summing over outgoing-photon helicities, we obtain the following relation

$$\begin{aligned} \frac{d\sigma}{d\Omega} &= \frac{e^4}{64\pi^2 m^2} \frac{\omega'^2}{\omega^2} \frac{1}{2} 4 \left[ \frac{\omega'}{\omega} + \frac{\omega}{\omega'} + \left[ 2 \cos^2 \frac{\vartheta}{2} + 2 \sin^2 \frac{\vartheta}{2} (1 - \delta_{hh'}) \right]^2 - 2 \right] \\ &= \frac{e^4}{32\pi^2 m^2} \frac{\omega'^2}{\omega^2} \left[ \frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2 \vartheta \right]. \end{aligned}$$

[5.] Finally, we use the low energy limit of the incoming photon  $\omega \ll m$ , which results in

$$\omega' = \frac{\omega}{1 + \frac{\omega}{m}(1 - \cos \vartheta)} \approx \omega,$$

obtaining the low energy limit differential cross-section

$$\left( \frac{d\sigma}{d\Omega} \right)_{\omega \ll m} = \frac{e^4}{32\pi^2 m^2} [1 + \cos^2 \vartheta].$$

Integrating this over  $d\Omega$  results in

$$\sigma_{\omega \ll m} = \frac{e^4}{32\pi^2 m^2} \int_0^{2\pi} d\varphi \int_{-1}^1 d\cos\vartheta [1 + \cos^2\vartheta] = \frac{e^4}{32\pi^2 m^2} 2\pi \left[2 + \frac{2}{3}\right].$$

Overall, we obtain

$$\sigma_{\omega \ll m} = \left(\frac{e^2}{4\pi m}\right)^2 \frac{8}{3}\pi.$$